

Stance Detection — vLLM Runtime & Carbon Summary

Dataset: SemEval 2016 Task 6 (n = 1,249) | Backend: vLLM | Models: 11 | Prompt variants: 11

Model	Prompt Variant	Runtime (s)	Runtime (min)	Total Tokens	Energy (kWh)	CO ₂ (g)
DeepSeek-R1-Distill-Llama-8B	default	204	3.4	638,868	0.018505	4.6876
DeepSeek-R1-Distill-Llama-8B	default_no_label_definitions	185	3.08	575,806	0.017941	4.5446
DeepSeek-R1-Distill-Llama-8B	question	67	1.12	683,073	0.008722	2.2093
DeepSeek-R1-Distill-Llama-8B	task_definition	81	1.35	665,023	0.01052	2.6647
DeepSeek-R1-Distill-Llama-8B	task_definition_scale	153	2.55	744,744	0.011133	2.8202
DeepSeek-R1-Distill-Llama-8B	few_shot	64	1.07	735,438	0.007688	1.9474
DeepSeek-R1-Distill-Llama-8B	few_shot_3	131	2.18	766,484	0.008492	2.151
DeepSeek-R1-Distill-Llama-8B	few_shot_6	52	0.87	946,832	0.006121	1.5506
DeepSeek-R1-Distill-Llama-8B	few_shot_9	67	1.12	1,126,626	0.005685	1.4401
DeepSeek-R1-Distill-Llama-8B	few_shot_12	49	0.82	1,316,058	0.005765	1.4604
DeepSeek-R1-Distill-Llama-8B	cot	111	1.85	1,253,986	0.016313	4.1324
DeepSeek-R1-Distill-Qwen-14B	default	303	5.05	581,095	0.027187	6.8868
DeepSeek-R1-Distill-Qwen-14B	default_no_label_definitions	246	4.1	516,993	0.02493	6.3151
DeepSeek-R1-Distill-Qwen-14B	question	98	1.63	623,718	0.013842	3.5064
DeepSeek-R1-Distill-Qwen-14B	task_definition	108	1.8	613,852	0.015298	3.8753
DeepSeek-R1-Distill-Qwen-14B	task_definition_scale	255	4.25	681,184	0.020529	5.2003
DeepSeek-R1-Distill-Qwen-14B	few_shot	92	1.53	698,606	0.012173	3.0836
DeepSeek-R1-Distill-Qwen-14B	few_shot_3	222	3.7	746,027	0.016027	4.0598
DeepSeek-R1-Distill-Qwen-14B	few_shot_6	69	1.15	916,004	0.008928	2.2617
DeepSeek-R1-Distill-Qwen-14B	few_shot_9	74	1.23	1,097,579	0.009173	2.3236

Model	Prompt Variant	Runtime (s)	Runtime (min)	Total Tokens	Energy (kWh)	CO ₂ (g)
DeepSeek-R1-Distill-Qwen-14B	few_shot_12	73	1.22	1,286,381	0.009568	2.4237
DeepSeek-R1-Distill-Qwen-14B	cot	166	2.77	1,200,541	0.025715	6.5139
Llama-3.1-8B-Instruct	default	91	1.52	304,225	0.006296	1.5948
Llama-3.1-8B-Instruct	default_no_label_definitions	32	0.53	252,959	0.002843	0.7202
Llama-3.1-8B-Instruct	question	24	0.4	332,016	0.002382	0.6034
Llama-3.1-8B-Instruct	task_definition	27	0.45	336,939	0.002528	0.6403
Llama-3.1-8B-Instruct	task_definition_scale	100	1.67	386,842	0.004615	1.169
Llama-3.1-8B-Instruct	few_shot	25	0.42	442,457	0.002435	0.6168
Llama-3.1-8B-Instruct	few_shot_3	94	1.57	498,706	0.004184	1.0599
Llama-3.1-8B-Instruct	few_shot_6	23	0.38	678,465	0.001513	0.3833
Llama-3.1-8B-Instruct	few_shot_9	21	0.35	849,529	0.001414	0.3581
Llama-3.1-8B-Instruct	few_shot_12	21	0.35	1,034,322	0.001397	0.354
Llama-3.1-8B-Instruct	cot	31	0.52	627,898	0.003727	0.9441
Nemotron-Mini-4B-Instruct	default	127	2.12	290,084	0.004548	1.152
Nemotron-Mini-4B-Instruct	default_no_label_definitions	16	0.27	235,227	0.000818	0.2072
Nemotron-Mini-4B-Instruct	question	16	0.27	312,712	0.000881	0.2231
Nemotron-Mini-4B-Instruct	task_definition	16	0.27	316,328	0.000872	0.2209
Nemotron-Mini-4B-Instruct	task_definition_scale	16	0.27	369,149	0.000947	0.2398
Nemotron-Mini-4B-Instruct	few_shot	16	0.27	493,857	0.0008	0.2025
Nemotron-Mini-4B-Instruct	few_shot_3	19	0.32	493,857	0.000855	0.2167
Nemotron-Mini-4B-Instruct	few_shot_6	15	0.25	676,144	0.000754	0.1911
Nemotron-Mini-4B-Instruct	few_shot_9	16	0.27	852,196	0.000829	0.2101

Model	Prompt Variant	Runtime (s)	Runtime (min)	Total Tokens	Energy (kWh)	CO ₂ (g)
Nemotron-Mini-4B-Instruct	few_shot_12	16	0.27	1,046,975	0.000786	0.199
Nemotron-Mini-4B-Instruct	cot	23	0.38	633,974	0.001721	0.4359
Qwen3-1.7B	default	42	0.7	282,245	0.003547	0.8986
Qwen3-1.7B	default_no_label_definitions	41	0.68	252,938	0.003245	0.8221
Qwen3-1.7B	question	33	0.55	362,698	0.002907	0.7363
Qwen3-1.7B	task_definition	35	0.58	319,781	0.00304	0.7701
Qwen3-1.7B	task_definition_scale	47	0.78	364,158	0.001984	0.5025
Qwen3-1.7B	few_shot	34	0.57	419,135	0.002634	0.6673
Qwen3-1.7B	few_shot_3	40	0.67	476,693	0.001692	0.4286
Qwen3-1.7B	few_shot_6	15	0.25	655,055	0.00074	0.1874
Qwen3-1.7B	few_shot_9	35	0.58	827,532	0.001196	0.303
Qwen3-1.7B	few_shot_12	15	0.25	1,013,631	0.000735	0.1861
Qwen3-1.7B	cot	57	0.95	1,110,163	0.006993	1.7713
Qwen3-14B	default	425	7.08	718,075	0.041575	10.5315
Qwen3-14B	default_no_label_definitions	421	7.02	664,210	0.0279	7.0674
Qwen3-14B	question	131	2.18	786,293	0.018778	4.7567
Qwen3-14B	task_definition	161	2.68	758,742	0.02156	5.4615
Qwen3-14B	task_definition_scale	238	3.97	1,221,188	0.035802	9.0691
Qwen3-14B	few_shot	146	2.43	864,465	0.019933	5.0493
Qwen3-14B	few_shot_3	284	4.73	916,579	0.021664	5.4877
Qwen3-14B	few_shot_6	171	2.85	1,091,803	0.017059	4.3212
Qwen3-14B	few_shot_9	112	1.87	1,268,266	0.01569	3.9745

Model	Prompt Variant	Runtime (s)	Runtime (min)	Total Tokens	Energy (kWh)	CO ₂ (g)
Qwen3-14B	few_shot_12	115	1.92	1,465,727	0.016151	4.0913
Qwen3-14B	cot	200	3.33	1,383,385	0.030626	7.7579
Qwen3-4B	default	151	2.52	742,510	0.015689	3.9741
Qwen3-4B	default_no_label_definitions	151	2.52	740,007	0.014837	3.7584
Qwen3-4B	question	80	1.33	868,059	0.010594	2.6836
Qwen3-4B	task_definition	79	1.32	820,564	0.01	2.533
Qwen3-4B	task_definition_scale	221	3.68	1,747,275	0.029171	7.3895
Qwen3-4B	few_shot	84	1.4	961,449	0.011263	2.8529
Qwen3-4B	few_shot_3	130	2.17	1,047,467	0.010388	2.6313
Qwen3-4B	few_shot_6	70	1.17	1,221,825	0.008688	2.2008
Qwen3-4B	few_shot_9	73	1.22	1,452,903	0.009303	2.3565
Qwen3-4B	few_shot_12	76	1.27	1,660,404	0.009753	2.4704
Qwen3-4B	cot	121	2.02	1,502,032	0.017067	4.3231
Qwen3-4B-Instruct-2507	default	56	0.93	277,049	0.004267	1.0808
Qwen3-4B-Instruct-2507	default_no_label_definitions	29	0.48	224,700	0.002576	0.6526
Qwen3-4B-Instruct-2507	question	21	0.35	303,639	0.002089	0.5291
Qwen3-4B-Instruct-2507	task_definition	22	0.37	309,723	0.001994	0.505
Qwen3-4B-Instruct-2507	task_definition_scale	60	1.0	358,597	0.002994	0.7585
Qwen3-4B-Instruct-2507	few_shot	21	0.35	413,525	0.00199	0.5041
Qwen3-4B-Instruct-2507	few_shot_3	82	1.37	472,290	0.003183	0.8063
Qwen3-4B-Instruct-2507	few_shot_6	28	0.47	652,138	0.001454	0.3683
Qwen3-4B-Instruct-2507	few_shot_9	21	0.35	824,497	0.001132	0.2868

Model	Prompt Variant	Runtime (s)	Runtime (min)	Total Tokens	Energy (kWh)	CO ₂ (g)
Qwen3-4B-Instruct-2507	few_shot_12	16	0.27	1,010,531	0.000991	0.251
Qwen3-4B-Instruct-2507	cot	36	0.6	745,400	0.004122	1.0442
Qwen3-8B	default	197	3.28	831,471	0.021189	5.3674
Qwen3-8B	default_no_label_definitions	144	2.4	790,313	0.011039	2.7963
Qwen3-8B	question	110	1.83	923,225	0.014094	3.5701
Qwen3-8B	task_definition	118	1.97	889,006	0.015637	3.961
Qwen3-8B	task_definition_scale	212	3.53	1,606,159	0.031553	7.9928
Qwen3-8B	few_shot	108	1.8	931,735	0.01098	2.7815
Qwen3-8B	few_shot_3	205	3.42	1,003,707	0.014769	3.7411
Qwen3-8B	few_shot_6	138	2.3	1,194,608	0.012968	3.285
Qwen3-8B	few_shot_9	78	1.3	1,361,120	0.010669	2.7026
Qwen3-8B	few_shot_12	86	1.43	1,592,162	0.011613	2.9417
Qwen3-8B	cot	151	2.52	1,511,152	0.022376	5.6682
phi-4	default	285	4.75	281,547	0.021196	5.3691
phi-4	default_no_label_definitions	278	4.63	230,385	0.023201	5.8772
phi-4	question	42	0.7	309,386	0.004453	1.1281
phi-4	task_definition	49	0.82	314,155	0.005289	1.3398
phi-4	task_definition_scale	171	2.85	361,831	0.008349	2.1148
phi-4	few_shot	43	0.72	415,519	0.00399	1.0108
phi-4	few_shot_3	159	2.65	471,680	0.00744	1.8847
phi-4	few_shot_6	31	0.52	651,631	0.002399	0.6076
phi-4	few_shot_9	35	0.58	822,756	0.002632	0.6667

Model	Prompt Variant	Runtime (s)	Runtime (min)	Total Tokens	Energy (kWh)	CO ₂ (g)
phi-4	few_shot_12	29	0.48	1,007,806	0.002315	0.5864
phi-4	cot	105	1.75	757,002	0.015155	3.8389
phi-4-mini-instruct	default	54	0.9	270,357	0.003434	0.8699
phi-4-mini-instruct	default_no_label_definitions	25	0.42	216,685	0.002377	0.6021
phi-4-mini-instruct	question	22	0.37	297,341	0.002165	0.5485
phi-4-mini-instruct	task_definition	22	0.37	301,693	0.001928	0.4883
phi-4-mini-instruct	task_definition_scale	65	1.08	346,967	0.002821	0.7146
phi-4-mini-instruct	few_shot	21	0.35	403,344	0.001719	0.4354
phi-4-mini-instruct	few_shot_3	58	0.97	459,480	0.002506	0.6348
phi-4-mini-instruct	few_shot_6	19	0.32	635,622	0.001049	0.2658
phi-4-mini-instruct	few_shot_9	18	0.3	804,263	0.000976	0.2474
phi-4-mini-instruct	few_shot_12	18	0.3	984,431	0.001003	0.2541
phi-4-mini-instruct	cot	31	0.52	632,815	0.00328	0.8308